

Initiating Search

November 6, 2025, 5:03 PM

• Search:

CAS Lexicon Search:

Lithium aluminum hydride - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (9,715)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (659)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%, 70-79%, 50-69%	
Document Type:	Journal	
Language:	English	

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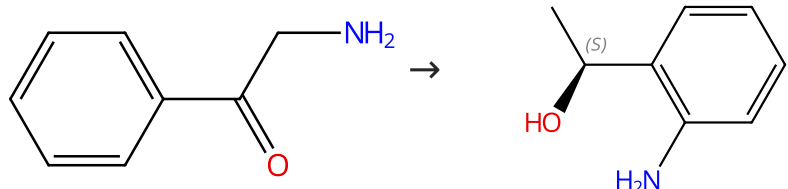
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Reactions (50)

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Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (33)
Absolute stereochemistry shown,
Rotation (+)
 Suppliers (22)

31-080-CAS-14872976

Steps: 1 Yield: 100%

A chiral bimetallic Lewis acid as a reaction template.

Asymmetric reduction of unsymmetric ketones with LiBH₄

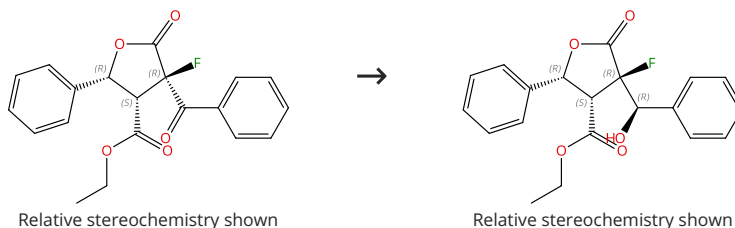
By: Nozaki, Kyoko; et al

Bulletin of the Chemical Society of Japan (1999), 72(5), 1109-1113.

- 1.1 Reagents: (4*R*,4'*R*)-2,2'-Bis[3,5-bis(trifluoromethyl)phenyl][4,4'-bi-1,3,2-dioxaborolane]-5,5'-dione
Solvents: Tetrahydrofuran
- 1.2 Reagents: Lithium borohydride
Solvents: Tetrahydrofuran
- 1.3 Reagents: Sodium hydroxide
Solvents: Water

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%



Relative stereochemistry shown

Relative stereochemistry shown

31-513-CAS-3569776

Steps: 1 Yield: 100%

Stereoselective Synthesis of 1-Fluoro-*exo,exo*-2,6-diaryl-3,7-dioxabicyclo[3.3.0]octanes: Synthesis of (±)-1-Fluoromembrine

By: Punirun, Teerachai; et al

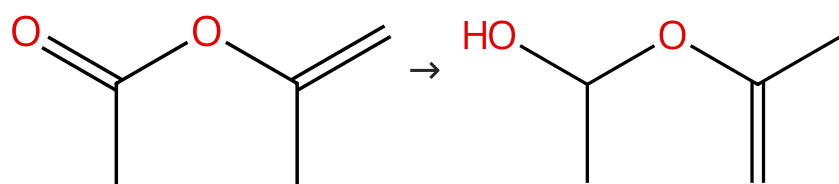
Journal of Organic Chemistry (2015), 80(16), 7946-7960.

- 1.1 Reagents: Diisobutylaluminum hydride
Solvents: Tetrahydrofuran, Hexane; 10 min, -78 °C; 1 h, -78 °C
- 1.2 Reagents: Potassium sodium tartrate
Solvents: Water; 0 °C

Experimental Protocols

Scheme 3 (1 Reaction)

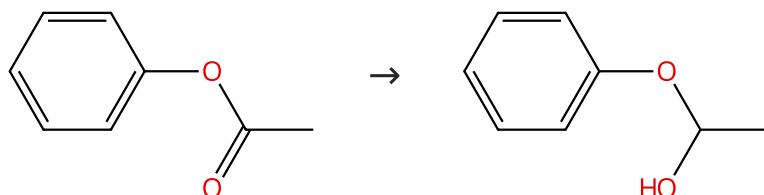
Steps: 1 Yield: 100%


 Suppliers (65)

31-513-CAS-8907904 Steps: 1 Yield: 100% 1.1 Reagents: Aluminate(1-), hydrotris(2-methyl-2-propanet hiolato)-, lithium (1:1), (7-4)- Solvents: Tetrahydrofuran; 1 h, 25 °C 1.2 Reagents: Hydrochloric acid Solvents: Water Experimental Protocols	Reaction of lithium tris(tert-butylthiolato)hydridoaluminate with selected organic compounds containing representative functional groups By: Cha, Jin Soon; et al Journal of Inclusion Phenomena and Macrocyclic Chemistry (2009), 65(1-2), 7-13.
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Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%



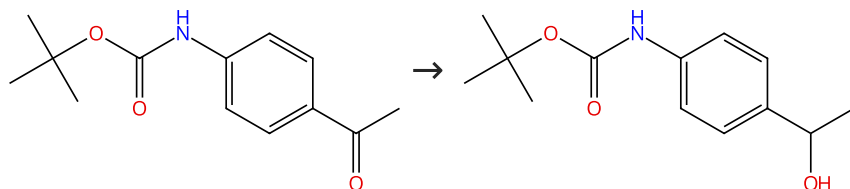
Suppliers (77)

Suppliers (4)

31-513-CAS-6774406 Steps: 1 Yield: 100% 1.1 Reagents: Aluminate(1-), hydrotris(2-methyl-2-propanet hiolato)-, lithium (1:1), (7-4)- Solvents: Tetrahydrofuran; 1 h, 25 °C 1.2 Reagents: Hydrochloric acid Solvents: Water Experimental Protocols	Reaction of lithium tris(tert-butylthiolato)hydridoaluminate with selected organic compounds containing representative functional groups By: Cha, Jin Soon; et al Journal of Inclusion Phenomena and Macrocyclic Chemistry (2009), 65(1-2), 7-13.
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Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (62)

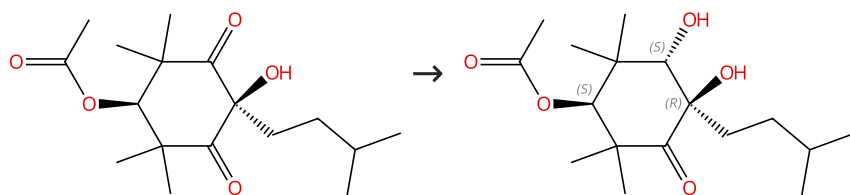
Rotation (+)

Suppliers (26)

31-513-CAS-7883770 Steps: 1 Yield: 100% 1.1 Reagents: (4<i>R</i>,4'<i>R</i>)-2,2'-Bis[3,5-bis(trifluoromethyl)phenyl][4,4'-bi-1,3,2-dioxaborolane]-5,5'-dione Solvents: Tetrahydrofuran 1.2 Reagents: Lithium borohydride Solvents: Tetrahydrofuran 1.3 Reagents: Sodium hydroxide Solvents: Water	A chiral bimetallic Lewis acid as a reaction template. Asymmetric reduction of unsymmetric ketones with LiBH₄ By: Nozaki, Kyoko; et al Bulletin of the Chemical Society of Japan (1999), 72(5), 1109-1113.
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Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%



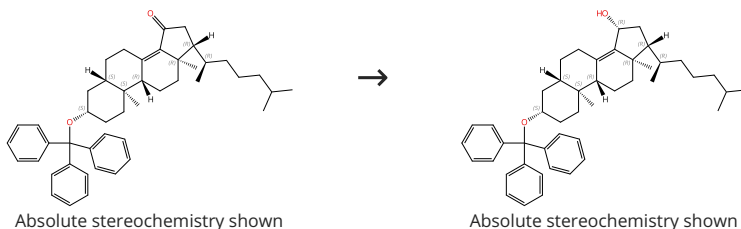
Relative stereochemistry shown

Relative stereochemistry shown

31-513-CAS-19246155 Steps: 1 Yield: 100% 1.1 Reagents: Sodium triacetoxyborohydride Solvents: Tetrahydrofuran; 18 h, rt 1.2 Reagents: Sodium chloride Solvents: Water; rt Experimental Protocols	Improved total synthesis of (±)-Tetragocarbone A By: Nishimura, Eiji; et al Tetrahedron (2018), 74(21), 2664-2668.
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Scheme 7 (1 Reaction)

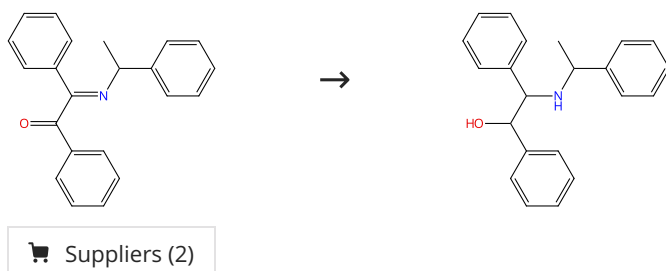
Steps: 1 Yield: 100%



31-513-CAS-7791736 Steps: 1 Yield: 100% 1.1 Reagents: Lithium aluminum hydride Solvents: Diethyl ether	Stereospecific reduction of keto group in 3β-triphenyl methoxy-5α-cholest-8(14)-en-15-one By: Ignatov, D. V.; et al Russian Journal of Bioorganic Chemistry (Translation of Bioorganicheskaya Khimiya) (2002), 28(2), 167-168.
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Scheme 8 (1 Reaction)

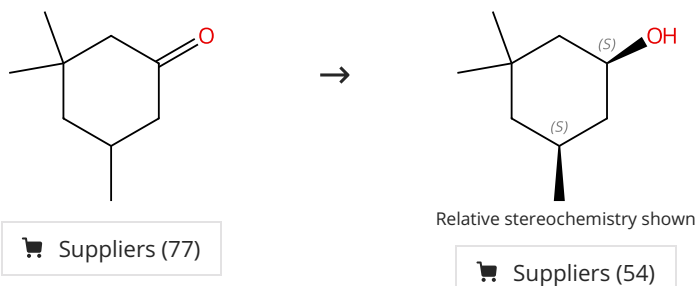
Steps: 1 Yield: 100%



31-513-CAS-4049758 Steps: 1 Yield: 100% 1.1 Reagents: Lithium aluminum hydride	The reduction of N-(1-phenylethyl)benzoylbenzylideneimine. A test for the estimation of the hard-soft character of reducing agents By: Alcaide, B.; et al Anales de Quimica, Serie C: Quimica Organica y Bioquimica (1982), 78(2), 278-9.
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Scheme 9 (1 Reaction)

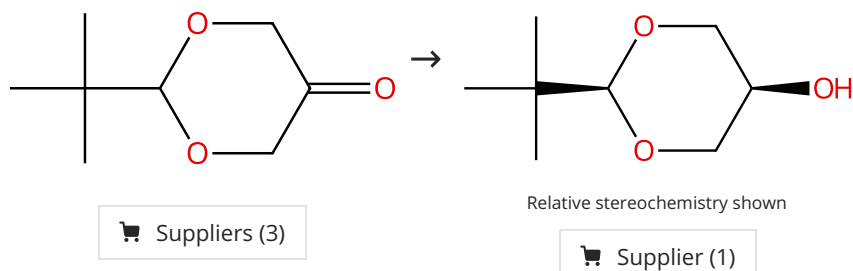
Steps: 1 Yield: 100%



31-513-CAS-14590867 Steps: 1 Yield: 100%	Reaction of sodium cyanoaluminum hydride with selected organic compounds containing representative functional groups By: Cha, Jin Soon; et al Bulletin of the Korean Chemical Society (2008), 29(12), 2379-2382.
1.1 Reagents: Aluminate(1-), (cyano-κC)trihydro-, sodium (1:1), (7-4)- Solvents: Tetrahydrofuran; 0.5 h, 0 °C 1.2 Reagents: Hydrochloric acid Solvents: Water	

Scheme 10 (2 Reactions)

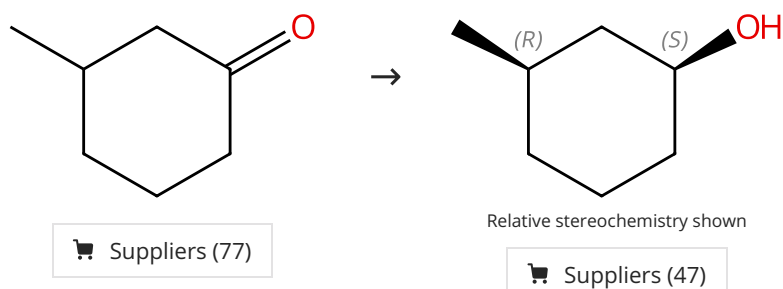
Steps: 1 Yield: 100%



31-513-CAS-1608826 Steps: 1 Yield: 100%	A high cis-stereoselectivity in the hydrogenation of 6-tert-butyltetrahydropyran-3-one and 2-tert-butyl-1,3-dioxan-5-one By: Senda, Yasuhisa; et al Bulletin of the Chemical Society of Japan (1989), 62(9), 2948-52.
1.1 Reagents: <i>p</i> -Toluenesulfonic acid, Hydrogen Catalysts: Palladium	
31-513-CAS-7700692 Steps: 1 Yield: 100%	A high cis-stereoselectivity in the hydrogenation of 6-tert-butyltetrahydropyran-3-one and 2-tert-butyl-1,3-dioxan-5-one By: Senda, Yasuhisa; et al Bulletin of the Chemical Society of Japan (1989), 62(9), 2948-52.
1.1 Reagents: <i>p</i> -Toluenesulfonic acid, Hydrogen Catalysts: Rhodium	

Scheme 11 (1 Reaction)

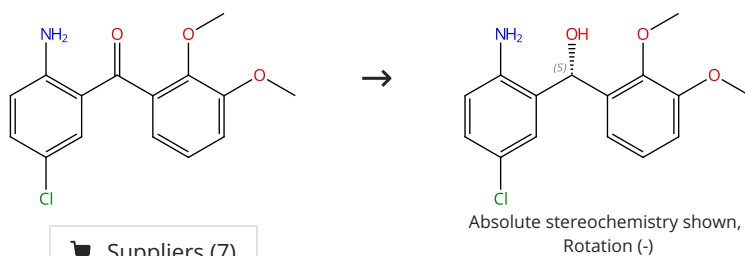
Steps: 1 Yield: 100%



31-513-CAS-6068943 Steps: 1 Yield: 100%	Reaction of sodium cyanoaluminum hydride with selected organic compounds containing representative functional groups By: Cha, Jin Soon; et al Bulletin of the Korean Chemical Society (2008), 29(12), 2379-2382.
1.1 Reagents: Aluminate(1-), (cyano-κC)trihydro-, sodium (1:1), (7-4)- Solvents: Tetrahydrofuran; 0.5 h, 0 °C 1.2 Reagents: Hydrochloric acid Solvents: Water	

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (7)

Suppliers (16)

31-513-CAS-3608193

Steps: 1 Yield: 100%

A chiral bimetallic Lewis acid as a reaction template.**Asymmetric reduction of unsymmetric ketones with LiBH₄**

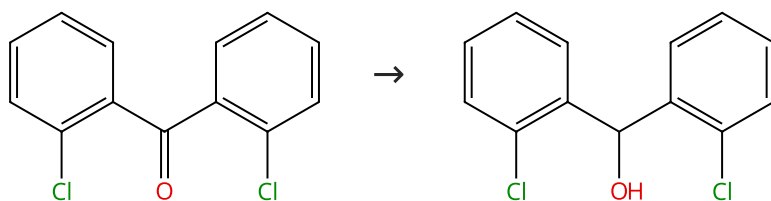
By: Nozaki, Kyoko; et al

Bulletin of the Chemical Society of Japan (1999), 72(5), 1109-1113.

- 1.1 **Reagents:** [4,4'-Bi-1,3,2-dioxaborolane]-5,5'-dione, 2,2-diphenyl-, [*R*-(*R*^{*},*R*^{*})]-
Solvents: Tetrahydrofuran
- 1.2 **Reagents:** Lithium borohydride
Solvents: Tetrahydrofuran
- 1.3 **Reagents:** Sodium hydroxide
Solvents: Water

Scheme 13 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (49)

Suppliers (47)

31-513-CAS-12861068

Steps: 1 Yield: 100%

LiAlH₄/silica chloride as a new chemoselective system for reduction of carbonyl compounds and phosphine oxides

By: Khalilzadeh, M. A.; et al

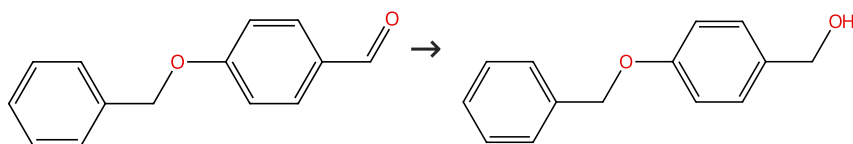
Journal of the Iranian Chemical Society (2008), 5(4), 699-705.

- 1.1 **Reagents:** Silica (chlorinated silica), Lithium aluminum hydride
Solvents: Diethyl ether; 12 min, rt

Experimental Protocols

Scheme 14 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (92)

Suppliers (82)

31-513-CAS-2219480

Steps: 1 Yield: 100%

LiAlH₄/silica chloride as a new chemoselective system for reduction of carbonyl compounds and phosphine oxides

By: Khalilzadeh, M. A.; et al

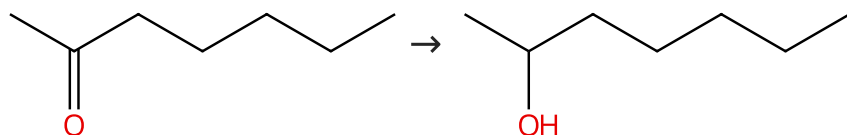
Journal of the Iranian Chemical Society (2008), 5(4), 699-705.

- 1.1 **Reagents:** Silica (chlorinated silica), Lithium aluminum hydride
Solvents: Diethyl ether; 15 min, rt

Experimental Protocols

Scheme 15 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (68)

Suppliers (77)

31-513-CAS-6398220

Steps: 1 Yield: 100%

Reaction of sodium cyanoaluminum hydride with selected organic compounds containing representative functional groups

By: Cha, Jin Soon; et al

Bulletin of the Korean Chemical Society (2008), 29(12), 2379-2382.

1.1 **Reagents:** Aluminate(1-), (cyano-κC)trihydro-, sodium (1:1), (7-4)-

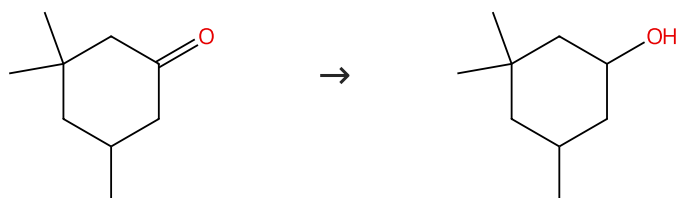
Solvents: Tetrahydrofuran; 0.5 h, 0 °C

1.2 **Reagents:** Hydrochloric acid

Solvents: Water

Scheme 16 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (77)

Suppliers (66)

31-513-CAS-7363896

Steps: 1 Yield: 100%

Reaction of lithium tris(tert-butylthiolato)hydridoaluminate with selected organic compounds containing representative functional groups

By: Cha, Jin Soon; et al

Journal of Inclusion Phenomena and Macrocyclic Chemistry (2009), 65(1-2), 7-13.

1.1 **Reagents:** Aluminate(1-), hydrotris(2-methyl-2-propanethiolato)-, lithium (1:1), (7-4)-

Solvents: Tetrahydrofuran; 1 h, 0 °C

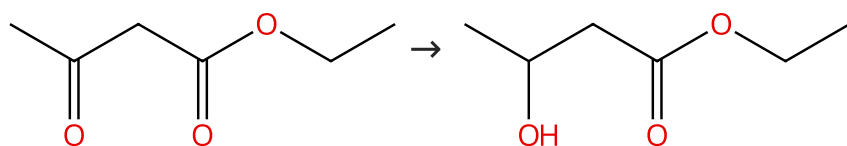
1.2 **Reagents:** Hydrochloric acid

Solvents: Water

Experimental Protocols

Scheme 17 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (95)

Suppliers (89)

31-513-CAS-13793494

Steps: 1 Yield: 100%

Glycerol as an alternative green medium for carbonyl compound reductions

By: Wolfson, Adi; et al

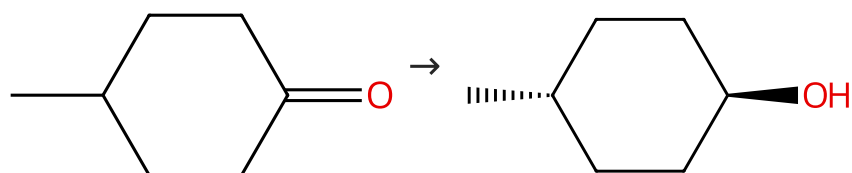
Organic Communications (2009), 2(2), 34-41.

1.1 **Reagents:** Sodium borohydride
Solvents: Glycerol; 0.5 h, 25 °C

1.2 **Reagents:** Sulfuric acid
Solvents: Water

Scheme 18 (1 Reaction)

Steps: 1 Yield: 100%



Relative stereochemistry shown

Suppliers (77)

Suppliers (63)

31-513-CAS-8205595

Steps: 1 Yield: 100%

Reaction of sodium cyanoaluminum hydride with selected organic compounds containing representative functional groups

By: Cha, Jin Soon; et al

Bulletin of the Korean Chemical Society (2008), 29(12), 2379-2382.

1.1 **Reagents:** Aluminate(1-), (cyano-κC)trihydro-, sodium (1:1), (7-4)-

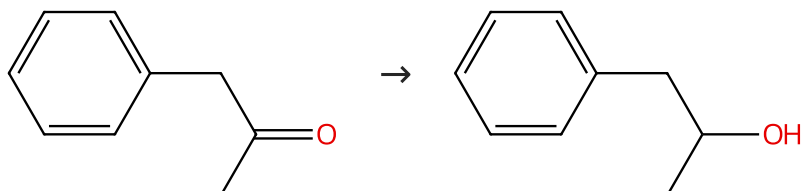
Solvents: Tetrahydrofuran; 0.5 h, 0 °C

1.2 **Reagents:** Hydrochloric acid

Solvents: Water

Scheme 19 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (26)

Suppliers (89)

31-513-CAS-8604279

Steps: 1 Yield: 100%

LiAlH₄/silica chloride as a new chemoselective system for reduction of carbonyl compounds and phosphine oxides

By: Khalilzadeh, M. A.; et al

Journal of the Iranian Chemical Society (2008), 5(4), 699-705.

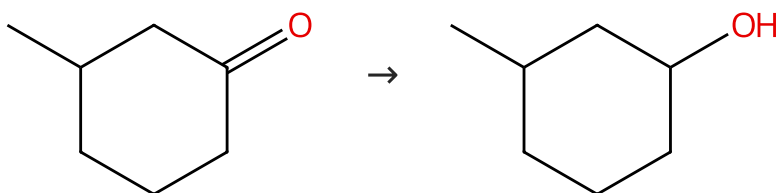
1.1 **Reagents:** Silica (chlorinated silica), Lithium aluminum hydride

Solvents: Diethyl ether; 20 min, rt

Experimental Protocols

Scheme 20 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (77)

Suppliers (51)

31-513-CAS-7092526

Steps: 1 Yield: 100%

Reaction of lithium tris(tert-butylthiolato)hydridoaluminate with selected organic compounds containing representative functional groups

By: Cha, Jin Soon; et al

Journal of Inclusion Phenomena and Macrocyclic Chemistry (2009), 65(1-2), 7-13.

1.1 **Reagents:** Aluminate(1-), hydrotris(2-methyl-2-propanethiolato)-, lithium (1:1), (7-4)-

Solvents: Tetrahydrofuran; 1 h, 0 °C

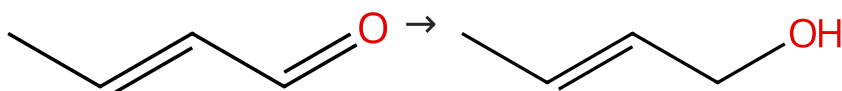
1.2 **Reagents:** Hydrochloric acid

Solvents: Water

Experimental Protocols

Scheme 21 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (14)

Suppliers (38)

31-513-CAS-704511

Steps: 1 Yield: 100%

Reaction of lithium tris(tert-butylthiolato)hydridoaluminate with selected organic compounds containing representative functional groups

By: Cha, Jin Soon; et al

Journal of Inclusion Phenomena and Macrocyclic Chemistry (2009), 65(1-2), 7-13.

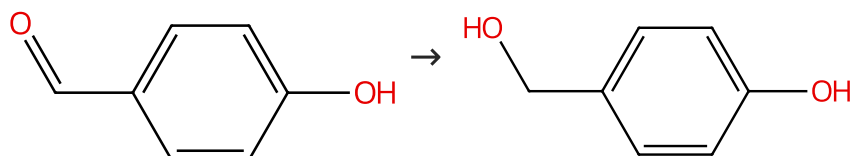
1.1 **Reagents:** Aluminate(1-), hydrotris(2-methyl-2-propanethiolato)-, lithium (1:1), (7-4)-
Solvents: Tetrahydrofuran; 24 h, 0 °C

1.2 **Reagents:** Hydrochloric acid
Solvents: Water

Experimental Protocols

Scheme 22 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (121)

Suppliers (113)

31-513-CAS-8936212

Steps: 1 Yield: 100%

LiAlH₄/silica chloride as a new chemoselective system for reduction of carbonyl compounds and phosphine oxides

By: Khalilzadeh, M. A.; et al

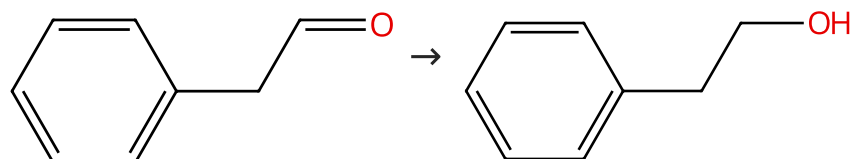
Journal of the Iranian Chemical Society (2008), 5(4), 699-705.

1.1 **Reagents:** Silica (chlorinated silica), Lithium aluminum hydride
Solvents: Diethyl ether; 25 min, rt

Experimental Protocols

Scheme 23 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (69)

Suppliers (110)

31-513-CAS-4671505

Steps: 1 Yield: 100%

LiAlH₄/silica chloride as a new chemoselective system for reduction of carbonyl compounds and phosphine oxides

By: Khalilzadeh, M. A.; et al

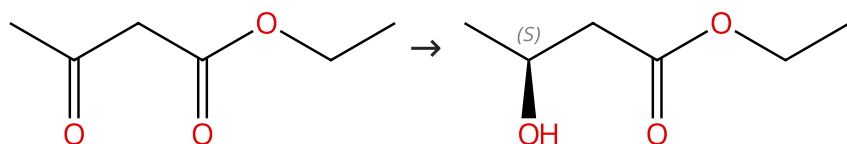
Journal of the Iranian Chemical Society (2008), 5(4), 699-705.

1.1 **Reagents:** Silica (chlorinated silica), Lithium aluminum hydride
Solvents: Diethyl ether; 10 min, rt

Experimental Protocols

Scheme 24 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (95)

Absolute stereochemistry shown,
Rotation (+)

Suppliers (72)

31-513-CAS-685448

Steps: 1 Yield: 100%

Glycerol as an alternative green medium for carbonyl compound reductions

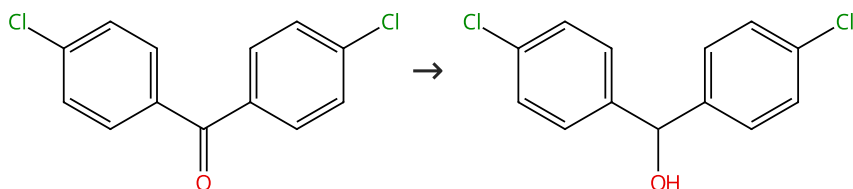
1.1 **Reagents:** Glucose
Solvents: Glycerol; 48 h, 37 °C

By: Wolfson, Adi; et al

Organic Communications (2009), 2(2), 34-41.

Scheme 25 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (80)

Suppliers (72)

31-513-CAS-14991598

Steps: 1 Yield: 100%

LiAlH₄/silica chloride as a new chemoselective system for reduction of carbonyl compounds and phosphine oxides

1.1 **Reagents:** Silica (chlorinated silica), Lithium aluminum hydride
Solvents: Diethyl ether; 12 min, rt

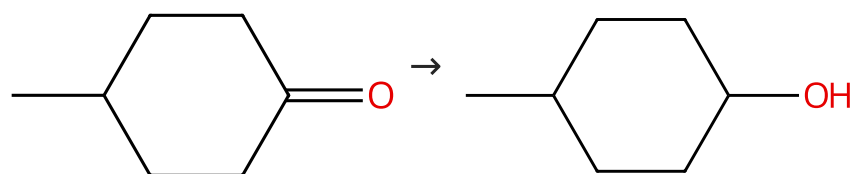
By: Khalilzadeh, M. A.; et al

Experimental Protocols

Journal of the Iranian Chemical Society (2008), 5(4), 699-705.

Scheme 26 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (77)

Suppliers (73)

31-513-CAS-9235384

Steps: 1 Yield: 100%

Reaction of lithium tris(tert-butylthiolato)hydridoaluminate with selected organic compounds containing representative functional groups

1.1 **Reagents:** Aluminate(1-), hydrotris(2-methyl-2-propanethiolato)-, lithium (1:1), (7-4)-
Solvents: Tetrahydrofuran; 1 h, 0 °C

By: Cha, Jin Soon; et al

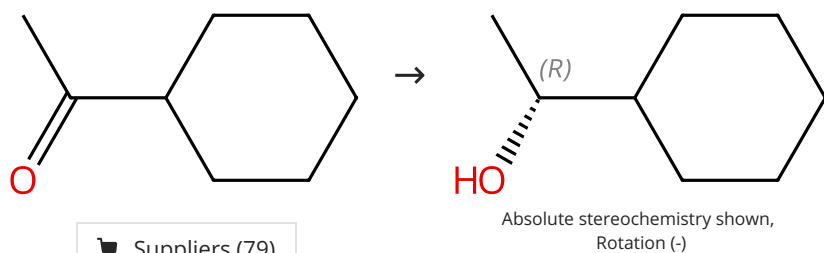
1.2 **Reagents:** Hydrochloric acid
Solvents: Water

Journal of Inclusion Phenomena and Macrocyclic Chemistry (2009), 65(1-2), 7-13.

Experimental Protocols

Scheme 27 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (79)

Suppliers (52)

31-513-CAS-1764969

Steps: 1 Yield: 100%

A chiral bimetallic Lewis acid as a reaction template.

Asymmetric reduction of unsymmetric ketones with LiBH_4

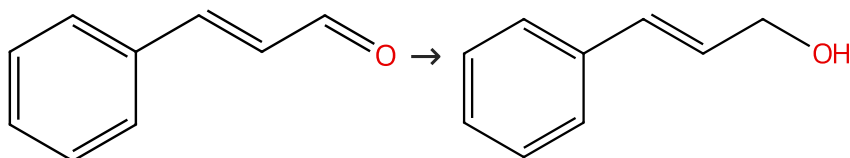
By: Nozaki, Kyoko; et al

Bulletin of the Chemical Society of Japan (1999), 72(5), 1109-1113.

- 1.1 **Reagents:** (4*R*,4'*R*)-2,2'-Bis[3,5-bis(trifluoromethyl)phenyl][4,4'-bi-1,3,2-dioxaborolane]-5,5'-dione
Solvents: Tetrahydrofuran
- 1.2 **Reagents:** Lithium borohydride
Solvents: Tetrahydrofuran
- 1.3 **Reagents:** Sodium hydroxide
Solvents: Water

Scheme 28 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (52)

Suppliers (86)

31-513-CAS-2853510

Steps: 1 Yield: 100%

Reaction of lithium tris(tert-butylthiolato)hydridoaluminate with selected organic compounds containing representative functional groups

By: Cha, Jin Soon; et al

Journal of Inclusion Phenomena and Macrocyclic Chemistry (2009), 65(1-2), 7-13.

- 1.1 **Reagents:** Aluminate(1-), hydrotris(2-methyl-2-propanethiolato)-, lithium (1:1), (7-4)-
Solvents: Tetrahydrofuran; 24 h, 0 °C
- 1.2 **Reagents:** Hydrochloric acid
Solvents: Water

Experimental Protocols

31-513-CAS-13466064

Steps: 1 Yield: 100%

 LiAlH_4 /silica chloride as a new chemoselective system for reduction of carbonyl compounds and phosphine oxides

By: Khalilzadeh, M. A.; et al

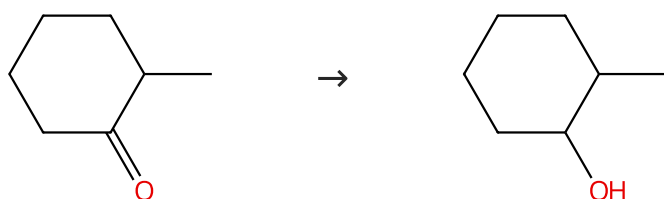
Journal of the Iranian Chemical Society (2008), 5(4), 699-705.

- 1.1 **Reagents:** Silica (chlorinated silica), Lithium aluminum hydride
Solvents: Diethyl ether; 15 min, rt

Experimental Protocols

Scheme 29 (2 Reactions)

Steps: 1 Yield: 100%



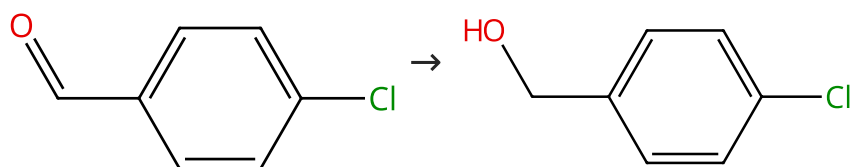
Suppliers (85)

Suppliers (71)

31-513-CAS-4969047 Steps: 1 Yield: 100% 1.1 Reagents: Aluminate(1-), hydrotris(2-methyl-2-propanethiolato)-, lithium (1:1), (7-4)- Solvents: Tetrahydrofuran; 1 h, 0 °C 1.2 Reagents: Hydrochloric acid Solvents: Water Experimental Protocols	Reaction of lithium tris(tert-butylthiolato)hydridoaluminate with selected organic compounds containing representative functional groups By: Cha, Jin Soon; et al Journal of Inclusion Phenomena and Macrocyclic Chemistry (2009), 65(1-2), 7-13.
31-513-CAS-3913752 Steps: 1 Yield: 100% 1.1 Reagents: Aluminate(1-), (cyano-κC)trihydro-, sodium (1:1), (7-4)- Solvents: Tetrahydrofuran; 0.5 h, 0 °C 1.2 Reagents: Hydrochloric acid Solvents: Water	Reaction of sodium cyanoaluminum hydride with selected organic compounds containing representative functional groups By: Cha, Jin Soon; et al Bulletin of the Korean Chemical Society (2008), 29(12), 2379-2382.

Scheme 30 (1 Reaction)

Steps: 1 Yield: 100%



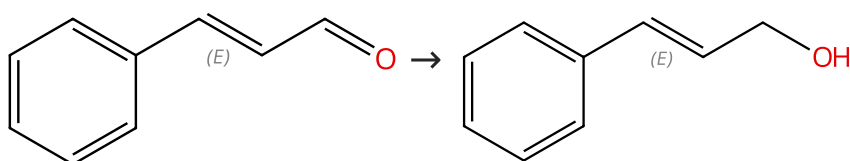
Suppliers (99)

Suppliers (85)

31-513-CAS-13187502 Steps: 1 Yield: 100% 1.1 Reagents: Silica (chlorinated silica), Lithium aluminum hydride Solvents: Diethyl ether; 6 min, rt Experimental Protocols	LiAlH₄/silica chloride as a new chemoselective system for reduction of carbonyl compounds and phosphine oxides By: Khalilzadeh, M. A.; et al Journal of the Iranian Chemical Society (2008), 5(4), 699-705.
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Scheme 31 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

Double bond geometry shown

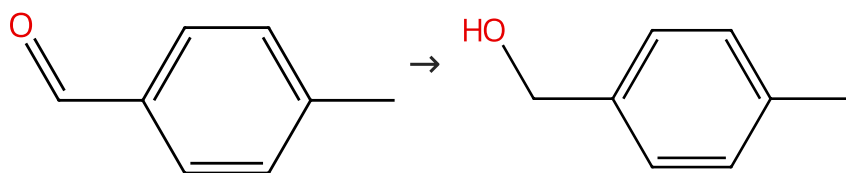
Suppliers (109)

Suppliers (66)

31-513-CAS-10952718 Steps: 1 Yield: 100% 1.1 Reagents: Aluminate(1-), (cyano-κC)trihydro-, sodium (1:1), (7-4)- Solvents: Tetrahydrofuran; 0.5 h, 0 °C 1.2 Reagents: Hydrochloric acid Solvents: Water	Reaction of sodium cyanoaluminum hydride with selected organic compounds containing representative functional groups By: Cha, Jin Soon; et al Bulletin of the Korean Chemical Society (2008), 29(12), 2379-2382.
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Scheme 32 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (104)

Suppliers (79)

31-513-CAS-4384567

Steps: 1 Yield: 100%

LiAlH₄/silica chloride as a new chemoselective system for reduction of carbonyl compounds and phosphine oxides

By: Khalilzadeh, M. A.; et al

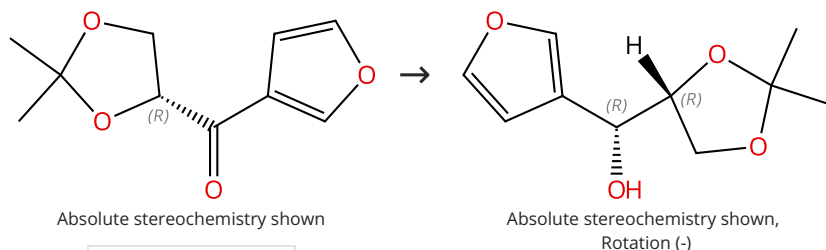
Journal of the Iranian Chemical Society (2008), 5(4), 699-705.

 1.1 **Reagents:** Silica (chlorinated silica), Lithium aluminum hydride
Solvents: Diethyl ether; 15 min, rt

Experimental Protocols

Scheme 33 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (2)

Supplier (1)

31-513-CAS-2015387

Steps: 1 Yield: 100%

On the diastereocontrol in the formation of (2R,3S)-3-(3'-furyl)-1,2-O-isopropylidenedioxy-3-pentanol and its (2R,3R)-diastereomer

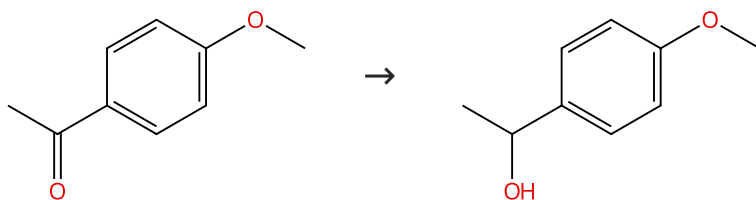
By: Hui, Chi Wai; et al

Tetrahedron Letters (2002), 43(1), 123-126.

 1.1 **Reagents:** Lithium triethylborohydride
Solvents: Tetrahydrofuran

Scheme 34 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (98)

Suppliers (66)

31-513-CAS-14667436

Steps: 1 Yield: 100%

LiAlH₄/silica chloride as a new chemoselective system for reduction of carbonyl compounds and phosphine oxides

By: Khalilzadeh, M. A.; et al

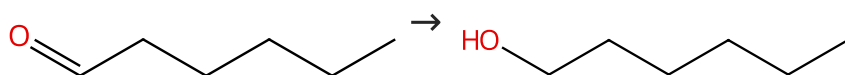
Journal of the Iranian Chemical Society (2008), 5(4), 699-705.

 1.1 **Reagents:** Benzoyl chloride, Silica (chlorinated silica), Lithium aluminum hydride
Solvents: Diethyl ether; 70 min, rt

Experimental Protocols

Scheme 35 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (68)

Suppliers (100)

31-513-CAS-11701778

Steps: 1 Yield: 100%

Reaction of lithium tris(tert-butylthiolato)hydridoaluminate with selected organic compounds containing representative functional groups

By: Cha, Jin Soon; et al

Journal of Inclusion Phenomena and Macrocyclic Chemistry (2009), 65(1-2), 7-13.

1.1 **Reagents:** Aluminate(1-), hydrotris(2-methyl-2-propanethiolato)-, lithium (1:1), (T-4)-
Solvents: Tetrahydrofuran; 1 h, 0 °C

1.2 **Reagents:** Hydrochloric acid
Solvents: Water

Experimental Protocols

31-513-CAS-4005310

Steps: 1 Yield: 100%

Reaction of sodium cyanoaluminum hydride with selected organic compounds containing representative functional groups

By: Cha, Jin Soon; et al

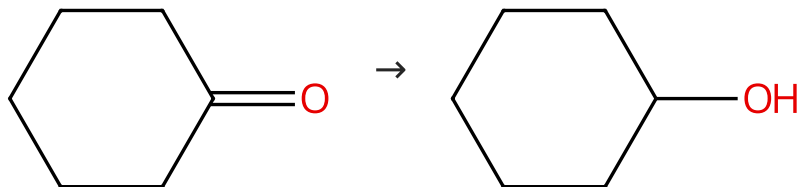
Bulletin of the Korean Chemical Society (2008), 29(12), 2379-2382.

1.1 **Reagents:** Aluminate(1-), (cyano-κC)trihydro-, sodium (1:1), (T-4)-
Solvents: Tetrahydrofuran; 0.5 h, 0 °C

1.2 **Reagents:** Hydrochloric acid
Solvents: Water

Scheme 36 (5 Reactions)

Steps: 1 Yield: 100%



Suppliers (102)

Suppliers (81)

31-513-CAS-8760980

Steps: 1 Yield: 100%

Applications of phase transfer catalysis. Part 48. A reevaluation of tetraalkylammonium tetrahydridoaluminate and phase transfer catalytic lithium aluminum hydride reductions

By: Dehmlow, Eckerhard V.; et al

Journal of Chemical Research, Synopses (1990), (1), 24.

1.1 **Reagents:** Lithium aluminum hydride
Catalysts: 2-(2-Methoxyethoxy)-N,N-bis[2-(2-methoxyethoxy)ethyl]ethanamine
Solvents: Toluene

31-513-CAS-10905656

Steps: 1 Yield: 100%

Applications of phase transfer catalysis. Part 48. A reevaluation of tetraalkylammonium tetrahydridoaluminate and phase transfer catalytic lithium aluminum hydride reductions

By: Dehmlow, Eckerhard V.; et al

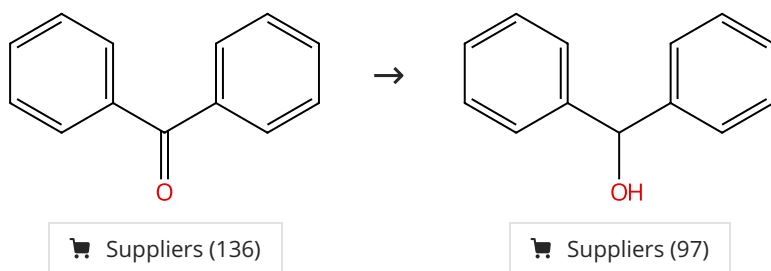
Journal of Chemical Research, Synopses (1990), (1), 24.

1.1 **Reagents:** Lithium aluminum hydride
Catalysts: Tetrahexylammonium bromide
Solvents: Toluene

31-513-CAS-6628964 Steps: 1 Yield: 100% 1.1 Reagents: Lithium aluminum hydride Catalysts: 15-Crown-5 Solvents: Toluene	Applications of phase transfer catalysis. Part 48. A reevaluation of tetraalkylammonium tetrahydridoaluminate and phase transfer catalytic lithium aluminum hydride reductions By: Dehmlow, Eckerhard V.; et al Journal of Chemical Research, Synopses (1990), (1), 24.
31-513-CAS-13017701 Steps: 1 Yield: 100% 1.1 Reagents: Lithium aluminum hydride Catalysts: Lithium iodide Solvents: Toluene	Applications of phase transfer catalysis. Part 48. A reevaluation of tetraalkylammonium tetrahydridoaluminate and phase transfer catalytic lithium aluminum hydride reductions By: Dehmlow, Eckerhard V.; et al Journal of Chemical Research, Synopses (1990), (1), 24.
31-513-CAS-4500397 Steps: 1 Yield: 100% 1.1 Reagents: Lithium aluminum hydride Catalysts: 18-Crown-6 Solvents: Toluene	Applications of phase transfer catalysis. Part 48. A reevaluation of tetraalkylammonium tetrahydridoaluminate and phase transfer catalytic lithium aluminum hydride reductions By: Dehmlow, Eckerhard V.; et al Journal of Chemical Research, Synopses (1990), (1), 24.

Scheme 37 (1 Reaction)

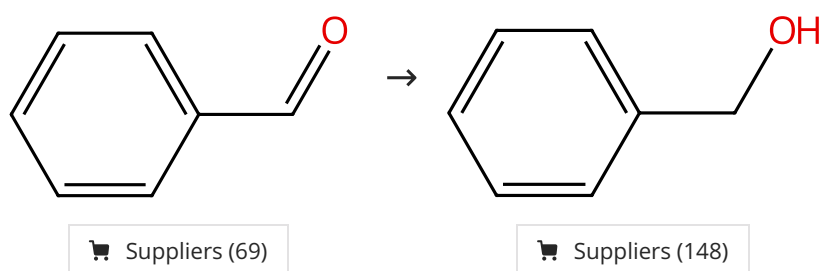
Steps: 1 Yield: 100%



31-513-CAS-12787470 Steps: 1 Yield: 100% 1.1 Reagents: Aluminate(1-), (cyano-κC)trihydro-, sodium (1:1), (7-4)- Solvents: Tetrahydrofuran; 0.5 h, 0 °C 1.2 Reagents: Hydrochloric acid Solvents: Water	Reaction of sodium cyanoaluminum hydride with selected organic compounds containing representative functional groups By: Cha, Jin Soon; et al Bulletin of the Korean Chemical Society (2008), 29(12), 2379-2382.
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Scheme 38 (5 Reactions)

Steps: 1 Yield: 100%



31-513-CAS-9037359 Steps: 1 Yield: 100% 1.1 Reagents: Sodium borohydride Solvents: Glycerol; 1 h, 25 °C 1.2 Reagents: Sulfuric acid Solvents: Water	Glycerol as an alternative green medium for carbonyl compound reductions By: Wolfson, Adi; et al Organic Communications (2009), 2(2), 34-41.
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31-513-CAS-2708832 Steps: 1 Yield: 100% 1.1 Reagents: Silica (chlorinated silica), Lithium aluminum hydride Solvents: Diethyl ether; 10 min, rt Experimental Protocols	LiAlH₄/silica chloride as a new chemoselective system for reduction of carbonyl compounds and phosphine oxides By: Khalilzadeh, M. A.; et al Journal of the Iranian Chemical Society (2008), 5(4), 699-705.
31-513-CAS-15526644 Steps: 1 Yield: 100% 1.1 Reagents: Aluminate(1-), (cyano-κC)trihydro-, sodium (1:1), (7-4)- Solvents: Tetrahydrofuran; 0.5 h, 0 °C 1.2 Reagents: Hydrochloric acid Solvents: Water	Reaction of sodium cyanoaluminum hydride with selected organic compounds containing representative functional groups By: Cha, Jin Soon; et al Bulletin of the Korean Chemical Society (2008), 29(12), 2379-2382.
31-513-CAS-15077971 Steps: 1 Yield: 100% 1.1 Reagents: Silica, Lithium aluminum hydride Experimental Protocols	Lithium aluminum hydride reduction in nonpolar solvents: use of silica gel as an effective catalyst By: Kamitori, Yasuhiro; et al Synthesis (1983), (5), 387-8.
31-513-CAS-6632839 Steps: 1 Yield: 100% 1.1 Reagents: Magnesium, [2,6-bis(1-methylethyl)phenolato] hydro- Solvents: Tetrahydrofuran	Reactions of magnesium hydrides. 1. Reduction of organic functional compounds by magnesium hydride and 2,6-diisopropylphenoxymagnesium hydride By: Ashby, E. C.; et al Journal of Organic Chemistry (1978), 43(8), 1557-60.